Fiscal Year:	FY 2023	**Task Last Updated:**	FY 07/13/2023
PI Name:	Friedman, Michael Ph.D.		
Project Title:	Musculoskeletal Response to Disuse in a Genetically Diverse Mouse Population		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mafriedman@vcu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	512-689-6670
Organization Name:	Virginia Commonwealth University		
PI Address 1:	401 West Main Street		
PI Address 2:	East Hall Room E1229		
PI Web Page:			
City:	Richmond	**State:**	VA
Zip Code:	23284	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Go for Launch
Start Date:	01/01/2022	**End Date:**	12/31/2022
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-GFL0003		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Go for Launch Prolonged spaceflight decreases bone and muscle volume, increasing injury risk. To investigate the role genetic diversity plays in these effects, we will use genetically unique diversity outbred (DO) mice. Male and female DO mice will have a limb immobilized for 3weeks. The results will inform future research by identifying diagnostic or therapeutic targets for a genetically diverse population.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:			

Task Progress:	Prolonged spaceflight decreases bone and muscle volume, increasing injury risk. To investigate the role genetic diversity plays in these effects, we used genetically uniqued iversity outbred (J:DO) mice. Male and female J:DO mice had a limb immobilized for 3 weeks. All mice had bone and muscle loss. Genetics affected the magnitude of bone and muscle loss. Female mice were more susceptible to bone loss than male mice. Despite extremely high variance, we were able to detect significant effects of immobilization. The results will inform future research by identifying diagnostic or therapeutic targets for a genetically diverse population. This is the first study using genetically unique mice to investigate the musculoskeletal response to unloading. We found strong evidence that the magnitude of bone and muscle loss is heritable. Bones in female mice were more susceptible to unloading than bones in male mice. J:DO mice can be used in future studies for space health research to discover genes regulating the musculoskeletal response to unloading and test new countermeasures using a more translatable mouse model.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Peer-reviewed Journals	Buettmann EG, Goldscheitter GM, Hoppock GA, Friedman MA, Suva LJ, Donahue HJ. "Similarities between disuse and age-induced bone loss." J Bone Miner Res. 2022 Jun 30;37(8):1417-34. Review. https://doi.org/10.1002/jbmr.4643 ; PMID: 35773785 , Jun-2023
Awards	Friedman M. "Alice L. Jee Young Investigator Award (and travel grant), June 2022." Jun-2023
Awards	Friedman M. "The Jackson Laboratory Diversity Outbred Pilot Grant Program, January 2022." Jan-2022